Disentangling emotions from the Web. Internet in the service of affect analysis

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Abstract

The paper presents a method for affect analysis of textual input in Japanese and its method of improvement by a Web mining technique. In the method, lexical emotive elements are cross-referenced with emotive expressions appearing in text. In the process of analysis, first a general emotiveness of an utterance is determined and then the specific types of conveyed emotional states are extracted. The system constructed on this method reached human level in determining general emotiveness and 63% of human level in extracting specific types of emotions. Applying the Web mining technique improved the performance of the emotional states types extraction to 85% of the human level.

Keywords: Affect analysis, emotiveness, Web mining, Human-Computer Interaction

1 Introduction

A rush in development of talking robots and conversational systems indicates that the functional implementation in our lives agents like intelligent car navigation systems [TW03] or talking furniture [HS07] has already become a current process, and there is a growing need for humanized robots fully understanding their users' cognitive behaviors.

One of the most important cognitive behaviors is expressing and understanding emotions. People make many decisions under the influence of their own emotions or the emotional states of others [Sc00, Yo06]. Therefore one of the most current issues of the fields like agent development or computer sciences is to produce methods for effective analyzing and understanding users' emotional states.

Recent introducing in our lives ubiquitous networks and wireless LAN made the Internet an indispensable "everyday article". Within the last few years there were several trials to retrieve from the Internet information concerning human emotions and attitudes [AC07]. However there are not many trials to use Web mining for the need of affect analysis. The affect analysis is focused generally on humanizing machines by implementing them methods for cognition of users' emotional states. It has became popular to analyze the emotions only from facial expressions or voice [KH00]. However, since emotions are strongly context dependent, most of the semantic content of expressing emotions is ignored in such researches.

Therefore we propose a method to analyze the affect of a textual input using the advantages of widely available today Internet resources for the need of real-time usage in robots interacting with humans.

2 Our Approach to Affect Analysis

Many attempts to analyze the emotive aspect of text end in a failure. Read [Re04] helplessly strived against the lack of appropriate emotive lexicon and the unsystemized classification of emotion types, achieving finally an accuracy of 33%[1]. Alm et al. [ARS05] achieved an accuracy of 69% and also met head on with the lacks in emotive databases, but also

[1] All results approximated to the nearest whole percent.

with an inappropriateness of corpus for evaluation consisting of children's stories full of ambiguities. Finally, Wu et al. [WC06] achieved 72% of general average accuracy, but run into a problem of ambiguity of emotional rules.

Therefore we made the following assumptions to specify the affect analysis. The research will be conducted on the basis of textual utterances. Comparing to large and quickly rising number of textual corpora, it is time consuming and uneconomical to gather a relevant corpus for the needs of analysis of vocal and visual dimensions. Moreover, neglecting the semantic meaning in such researches leads to ambiguities. Restricting to a textual corpus gives thus a green light to concentrate on emotiveness realized in the semantic surface of text. The analysis will be conducted on dialogue-like utterances appearing usually in speech. This approach relates to the future goal, which is to implement the system into a conversation system like Higuchi's [HRA08] to make its utterance understanding and generation more natural. The analysis of affect is performed in three steps: 1. Determining the emotiveness of an utterance, or finding whether an utterance is emotive or not; 2. Finding how strong are the emotions conveyed in the utterance, or setting the emotive value; 3. If the sentence is emotive, finding the types of emotions conveyed in the utterance.

2.1 Emotiveness - Linguistic Approach

In an everyday conversation there are different linguistic and paralinguistic means used to inform interlocutors about emotional states. Elements of speech used to convey emotive meaning in the sentence make up the feature of language called emotiveness [St63] and the function of language describing them is called the emotive function of language [Bu34, Ja60].

The emotive function of language is realized lexically in the Japanese through such parts of speech as the following: exclamations [Be02, On02], hypocoristics (endearments) [PK04], vulgar language [Cr89] or *gitaigo* - mimetic expressions [Ba03]. A key role in expressing emotions is also played by the lexicon of words describing states of emotions [Na93]. The analysis of elements of language such as intonation or tone of voice as well as nonverbal elements, like gestures or face expressions, in applications where the communication channel is limited to transmission of lexical symbols has to be focused on its textual and lexical manifestations, like exclamation marks ellipsis, or face-marks.

3 Definition and Classification of Emotions

We use the general definition of emotions proposed by Nakamura [Na93], who defines them as every temporary state of mind, feeling or emotional state evoked by experiencing different sensations. This definition is complemented by Beijer's [Be02] defining emotive utterances as every utterance in which the speaker is emotionally involved, and this involvement is linguistically expressed.

Nakamura [Na93], after a thorough study on emotions in the Japanese, proposes a fully systemized and coherent classification of emotions into 10 types - said to be the most appropriate for the language. That is: *ki / yorokobi* (joy, delight), *do / ikari* (anger), *ai / aware* (sorrow, sadness), *fu / kowagari* (fear), *chi / haji* (shame, shyness, bashfulness), *kou / suki* (liking, fondness), *iya / iyodomi* (dislike, detestation), *kou / takaburi* (excitement), *an / yasuki* (relief) and *kyou / odoroki* (surprise, amazement).

4 ML-Ask

Basing on the linguistic approach and assumptions described above, Ptaszynski et al. [PD08] constructed ML-Ask (Emotive Elements / Emotive Expressions Analysis System) for analyzing emotive content of utterances. The system performs three-step procedure: 1) Determining the emotiveness of an utterance; 2) Calculating the emotive value representing the strength of the conveyed emotions; 3) Recognizing the specific emotion types. The system is based on Ptaszynski's [Pt06] idea of binary classification of realizations of emotions in language into:
1. *Emotive elements*. Informing that emotions have been conveyed, but do not expressing specified feelings or expressing different ones, depending on the context. Examples are: *sugee* (great!), *wakuwaku* (heart pounding), *-yagaru* (fu**ing do sth);
2. *Emotive expressions*. Parts of speech, that in emotive sentences describe emotional states. Examples are: *aijou* (love), *kanashimu* (feel sad), *ureshii* (happy).

If the system discovers emotive elements, the utterance is marked as emotive. In emotive utterances ML-Ask searches for emotive expressions to precise the type of emotion conveyed.

The emotive elements databases were gathered from different research [OT95-8, Ts99, Ba03, Sj06] and divided into interjections, mimetics, endearments, vulgarities and representations of non-verbal emotive elements, like exclamation marks or ellipsis. Also there was added an algorithm detecting emoticons, as they are symbols commonly used in every-

day text-based communication tools. The databases of emotive expressions were created on Nakamura's [Na93] collection. This basic extraction procedure (see Fig. 1) is upgraded with a Web mining technique described below in detail.

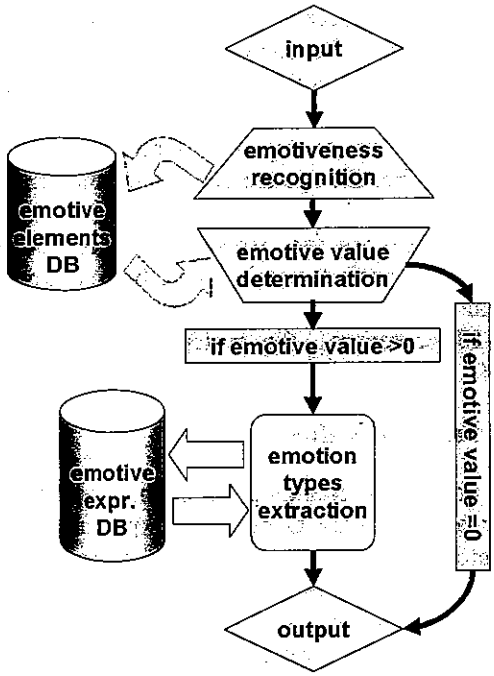


Fig. 1. Flow chart of the ML-Ask system.

5 Web Mining Technique

To improve the extraction of the specific emotion types we apply Shi's Web mining technique for extracting emotive content from the Web [SR08] upgraded with an automatic phrase extraction method based on dividing the sentence into separate n-gram phrases.

Original utterance	*Aa, pasokon ga kowarete shimatta...* (Darn, the PC has broken...)	
(n-1)-gram A	*Aa, pasokon ga kowareru*	
(n-1)-gram B	*pasokon ga koware te shimau*	
⋮	⋮	
3-grams	*pasokon ga kowareru*	*koware te shimau*

Table 1. Example of n-gram separating from an utterance.

5.1 Phrase Extraction From an Utterance

In an utterance processed morphologically every separated part of speech is treated as a 1-gram. All the 1-grams are grouped into larger n-gram groups preserving their word order in the utterance. The groups are arranged from the longest n-gram (the whole sentence) to all the groups of 3-grams, excluding those ending with particles, since they gave too many ambiguous results during test phase. An example of such n-gram separating procedure is shown in Table 1.

	Original n-gram	*pasokon ga koware te shimau*
n-gram phrase adjusting (morpheme modification)	*/ -te /*	*pasokon ga koware te shima- tte*
	/ -to /	*pasokon ga koware te shimau to*
	/ -node /	*pasokon ga koware te shimau node*
	/ -kara /	*pasokon ga koware te shimau kara*
	...	...

Table 2. Examples of n-gram modifications for Web mining.

5.2 Emotion Types Extraction From the Web

In the web mining technique the extracted phrase is used as a query in Google search engine. The n-grams ending with a verb or an adjective are adjusted for semantically deeper Web mining (See Table 2.), according to a thesis on causality morphemes conveying emotive meaning [Ya99], confirmed experimentally by Shi [SR08]. The morpheme modification provides additional query phrases. All the phrases are queried in Google, and a maximum to 500 snippets for each are extracted and cross-referenced with the databases of emotive expressions. The quantity of extracted emotive expressions belonging to a precise type of emotion provides a list of emotions commonsensically associating with the queried sentence.

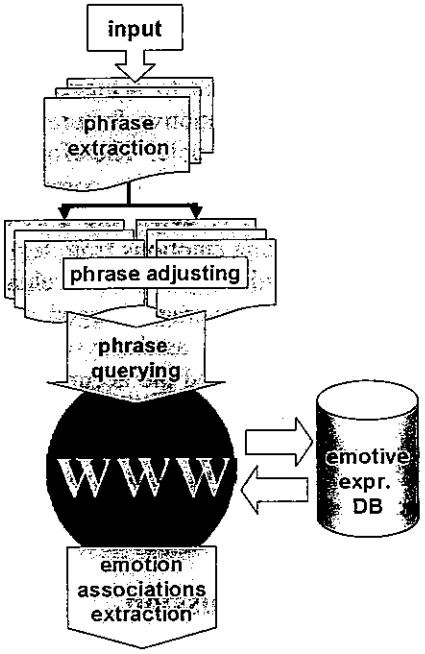


Fig. 2. Flow chart of the Web mining technique.

6 Evaluation Experiment

To verify the system's accuracy we performed an evaluation experiment. In the experiment we evaluated the ML-Ask system's performance for the three steps of the procedure described in section 5 and compared results for emotion types extraction for three different extraction procedures: 1) ML-Ask basic procedure; 2) ML-Ask with Shi's technique as a supportive mean; 3) ML-Ask with Shi's technique as the only mean for emotions extraction. For each extraction procedure we also checked the two variants mentioned in section 6.2 – taking into consideration all of the extracted emotions; and leaving only the ones with 3 best results considering the rest as a noise.

The evaluation is based on a corpus of natural utterances gathered through an anonymous survey. In the survey participated 30 people from different age and social groups. Each of them was to imagine or remember a conversation (or conversations) with any person (or persons) they know and write three sentences from that conversation: one free, one emotive, and one non-emotive. After that the sentences' authors tagged the utterances written by themselves in the same way, as the system's procedure - first it was determined whether an utterance is emotive. If it is, they set the emotive value (0-5) and describe the specific emotion types conveyed in emotive utterances.

We used ML-Ask to analyze this corpus of 90 sentences, and compared the results of the system with the results provided by the authors of the utterances. Although the ideal would be a perfect unanimity, it is highly difficult to achieve such score, since determining about emotive features of speech differs greatly between people. Therefore, we also added an emotive tagging of the corpus by third party human evaluators to determine a general human level in recognizing emotions from text in ordinary people. The third party human evaluators (10 people per sentence on average) tagged the corpus emotively in the same way as the system does and for each of them we counted the accuracy in recognition by comparing the results to the results provided by the authors of the utterances. The scope of results achieved by the third party evaluators was considered as human level of accuracy in recognizing emotions from text.

7 Results and Discussion

7.1 Emotive / Non-emotive

The system's accuracy in determining the emotiveness is calculated as a typical balanced F-score separately for emotive and non-emotive utterances.

The total accuracy of ML-Ask in determining general emotiveness evaluated on 90 items reached 0.83 of the balanced F-score. The result is very promising, since the same value checked for the third party human evaluators gave a wide range of results from 0.4 to 0.86. The result of ML-Ask was placed closely to the top of this ranking, therefore we can say that the system recognizes the emotiveness on a very high human level.

As an interesting fact, we should add that for sentences with a perfect match, where both authors and all evaluators were unanimous about the emotiveness (18 sentences), the system's results reached F=1.00.

7.2 Emotive Value

Many constantly changing situational features of a conversation as well as personal experiences of interlocutors make the emotive value of an utterance difficult to set with a strict precision.

Since the F-score is thus inappropriate for this stage of evaluation, we apply a coefficient of unanimity U between one evaluator (or author of an utterance) Eva_A and the other Eva_B (or system) counted as a relationship of the number of similarly tagged utterances t_{sim} to all of the utterances tagged by the human evaluators t_i, as in equation 1.

$$U_{Eva_B}^{Eva_A} = \frac{t_{sim}}{t_i} \tag{1}$$

We also assent to a condition of almost-perfect match, when the emotive value differs between the author of an utterance and the system by ±1 emotive point.

For 30 items available for this stage of evaluation, U reached 0.53. The accuracy in setting the emotive value in the third party showed a tendency of U=0.84. The system's result reaches 63% of the human level accuracy, thus it is desirable both, to find a more precise way of setting the emotive value in the future than the quantitative one used this time, as well as to increase the number of third party evaluators to increase the objectivity of the judgment. However, we consider the result as promising, since the emotional intensity setting is highly subjective among people. It is also predictable that upgrading the accuracy of determining the emotiveness of a sentence by eliminating the ambiguities in emotional affiliations of emotive elements will positively influence the emotive value setting accuracy.

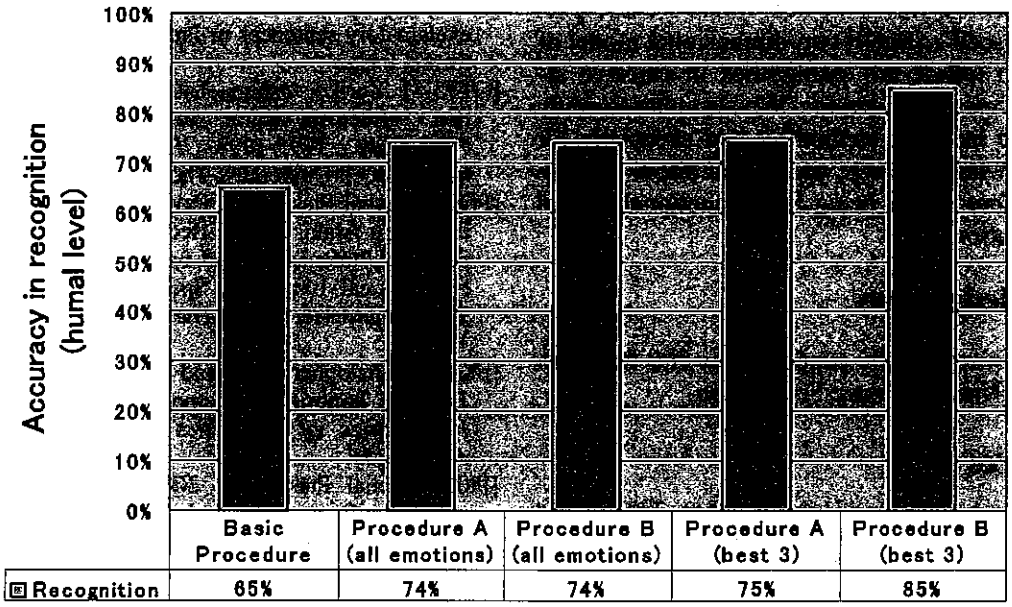


Fig. 3 Summarized results for 3 different procedures of emotion types extraction with 2 variants of procedures using Web mining technique.

7.3 Specified Emotions

By recognizing a specified type of emotion we understand a result of recognizing any- and at least one feeling from the utterance, including non-emotive (the results are summarized in Fig. 3).

7.3.1 Basic Procedure

The system's accuracy in estimating the specific types of emotions reached of 0.45 of the balanced F-score. As for human evaluators, the average accuracy was set at a level of 0.72 of balanced F-score. Therefore the system's accuracy is set at approximately 63% of the human level of emotions recognition. In the future research it is desirable to both - upgrade the algorithm for finding specified emotion types and perform the evaluation on larger number of tagged utterances.

7.3.2 Procedure A

This way of emotion types extraction uses the Web mining technique as a supportive mean for ML-Ask, where the system was not able to specify the emotion types. ML-Ask supported with this method achieved a balanced F-score of 0.54 and 0.53 for the two variants – "3 best results" and "all emotions" respectively. This gives approximately 75% and 74% of the human level in determining the emotion types.

7.3.3 Procedure B

In this way of emotion types extraction, the Web mining technique is used instead of the system's lexical extraction. ML-Ask with this method achieved respectively 0.61 and 0.53 of balanced F-score for the two variants. Human level in determining the types of emotions amounted then 85% and 74% respectively.

8 Conclusions and Future Work

In this paper we presented a method for affect analysis of utterances, supported with a Web mining technique. In every case the Web mining improved the accuracy of the system in extracting the specific types of emotions. There is a need for further experiments on a larger evaluation material to determine the most accurate extraction procedure. Experiments with different variants of the method showed, that it is more effective to keep only the emotions that achieved the three best results and consider the rest as a noise, rather than to take into consideration all of the extracted emotions. Furthermore, since the system's procedures are activated gradually, improving accuracy in earlier stages (emotiveness determination, emotive value calculation) clearly prognosticates improving of the emotion types extraction. The tools for morphological analysis used in the system are also not perfect, which might decrease the system's real accuracy. However improving the extraction of n-gram phrases from the queried sentence will surely elimi-

nate these difficulties. In the near future we also plan to apply Russell's [Ru80] two-dimensional model of emotions to reduce the ambiguities in emotive affiliations of emotive elements. We also plan to apply a Web page indexing method created especially for the needs of this research, which will allow the system to perform faster being fully functional in real time applications.

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